Short note for Joanna

Offline Rowing IMU Dashboard - current workflow and segmentation update

Hi Joanna,

I think the offline dashboard is now much closer to what we wanted for coach-facing analysis. I also finally understood an important point that was probably clear from the beginning: stroke detection should not mainly be based on peak acceleration. The current version now uses the change into positive seat velocity to detect the start of the drive, and it defines strokes from one drive start to the next drive start.

Why this makes more sense:

- Peak acceleration can be affected by vibration, bumps, or short unstable movements.
- Seat velocity is a better signal for identifying when the seat actually starts moving through the drive.
- The stroke window is now drive start to next drive start, instead of something closer to acceleration peak to acceleration peak.
- This should make the report more intuitive for coaches, because the detected stroke window follows the movement pattern rather than just the strongest acceleration spike.

Recommended practical workflow:

- Connect or power on both IMUs before the session: one device should be configured as SEAT, the other as BOAT.
- Start both devices as close together as practical. They do not need perfect hardware synchronization, but starting them close together makes the later time notes easier to use.
- Let athletes row one after another without stopping the recording.
- Write down approximate time windows from the start of the recording, for example Athlete 1: 00:10-01:15, Athlete 2: 01:30-02:40.
- After the session, copy both CSV files into one folder and generate the HTML report from that folder.
- Inside the HTML report, segment the recording by athlete time windows and inspect/export the athlete-specific views.

Synchronization note:

The two microcontrollers are not hardware-synchronized. For offline analysis, the report uses the recorded sample timing and compares SEAT and BOAT over the overlapping time range. This is sufficient for the current dashboard goal: coach-friendly comparison, segmentation, and visual analysis. For stricter scientific timing, a later version could add a shared synchronization event or hardware sync.

How to create the report:

Run the Python offline analysis script on the folder that contains both CSV files. The output is a self-contained HTML report, so it can be opened and shared without Python afterwards.

```
/usr/local/bin/python3 sketch_mar25a/analyze_log.py /path/to/folder_with_csv_files
--output rowing_report.html --label rowing_report --min-peak-ms 800 --smooth-window 10
```

My hope is that the dashboard is now much more understandable for coaches: record once, segment athletes afterwards, inspect stroke timing and boat response, and only tune the detection settings if the counted strokes do not match the visible/manual count.